



WIRE & CABLE FOR THE GAS POWER CAMPUS

THE PATH OF POWER

From the gas yard to the 230 kV line, every megawatt a plant makes travels by cable. This is the route, and the cable it runs on.

ACCELERATING TIME TO POWER™

SOUTHWIRE • POWER GENERATION SOLUTIONS

EVERY MEGAWATT TRAVELS BY CABLE.

Gas arrives at the yard and burns in three H-class turbines. Each generator makes power at 21 kV; a generator breaker and isolated-phase bus carry it to the step-up transformer, and 230 kV cable and conductors take it to the grid. Along the way, miles of medium- and low-voltage cable run every pump, fan and valve that keeps the turbines turning.

4.6 MI

of cable-tray routes

4.2 MI

of duct-bank and buried cable routes

417

pieces of equipment, each with its own cable

GAS TURBINE + GENERATOR
21 kV

STEP-UP TRANSFORMER
21 / 230 kV

TO THE GRID
230 kV

FUEL GAS

FUEL GAS

GENERATION · 21 kV

AUXILIARIES · 15 kV / 600 V

STEP-UP · 21 / 230 kV

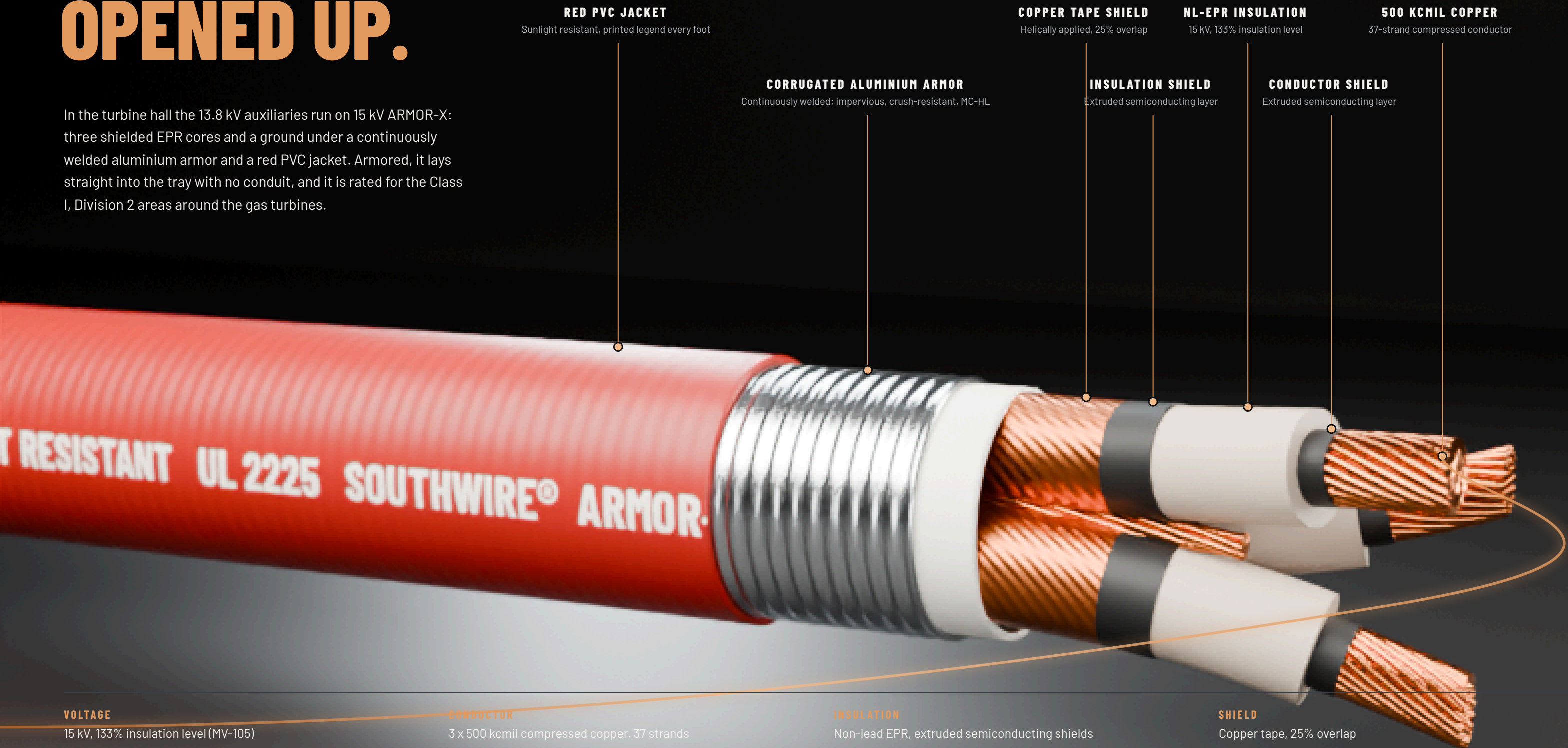
GRID · 230 kV

SK-3X1 campus, rendered from the 3D model. Route lengths measured along modelled routes, not cable footage.

ARMOR-X[®] OPENED UP.

In the turbine hall the 13.8 kV auxiliaries run on 15 kV ARMOR-X: three shielded EPR cores and a ground under a continuously welded aluminium armor and a red PVC jacket. Armored, it lays straight into the tray with no conduit, and it is rated for the Class I, Division 2 areas around the gas turbines.

Typical construction from the customer's ARMOR-X list (500-37 3/C CPRESS CU NL-EPR 25% T/S 1X#1 CU GW, red PVC). Confirm against current Southwire specifications. Studio render of the cable model, not a photograph.



VOLTAGE 15 kV, 133% insulation level (MV-105)	CONDUCTOR 3 x 500 kcmil compressed copper, 37 strands	INSULATION Non-lead EPR, extruded semiconducting shields	SHIELD Copper tape, 25% overlap
GROUND 1 x #1 AWG bare copper	ARMOR Continuously welded corrugated aluminium	JACKET Red PVC, sunlight resistant	LISTING Type MC-HL (UL 1569 / UL 2225) or MV-105 (UL 1072)
FUEL GAS	GENERATION - 21 / 230 kV	AUXILIARIES - 15 kV / 600 V	STEP-UP - 21 / 230 kV
			GRID - 230 kV